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TREND-BASED MOCK PAPER

GATE Mathematics

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mathematics
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Mathematics

Q1. [1-NAT] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q2. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q3. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q4. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q6. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-NAT] [1 mark]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q9. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u,v \rangle| \leq \sqrt{\langle u,u \rangle \langle v,v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q10. [1-NAT] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
 - (B) 0
 - (C) -1
 - (D) $\cos(\pi)$
-

Q12. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q13. [1-MCQ] [1 mark]

Laplace transform of $e^{-3t} \cdot u(t)$ is:

- (A) $1/(s+3)$
 - (B) $1/(s-3)$
 - (C) $1/s$
 - (D) e^{-3}/s
-

Q14. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q15. [1-NAT] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q17. [1-NAT] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
 - (B) $2\pi r$
 - (C) πr
 - (D) $2\pi r^2$
-

Q21. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
 - (B) -2
 - (C) 1
 - (D) 2
-

Q24. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MSQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q26. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q27. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q28. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q29. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q31. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

The Laplace transform of $t e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q33. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q34. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q36. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q37. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q38. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q39. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q40. [1-NAT] [1 mark]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q41. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q42. [2-MSQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q43. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q45. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q46. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q47. [2-NAT] [2 marks]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q49. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q51. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q52. [2-MCQ] [2 marks]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \cdot 2$
 - (D) $P \cdot r/100$
-

Q53. [2-MSQ] [2 marks]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \cdot 2$
 - (D) $P \cdot r/100$
-

Q54. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q56. [1-NAT] [1 mark]

Rank of 3×3 identity:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Ring must have:

- (A) 1 only

- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q58. [1-MCQ] [1 mark]

Inverse of $AB = ?$

- (A) $B^{-1} A^{-1}$
- (B) $A^{-1} B^{-1}$
- (C) $(A+B)^{-1}$
- (D) None

Q59. [1-MCQ] [1 mark]

Identity element property:

- (A) $a * e = e * a = a$
- (B) $a * e = e$
- (C) $e * a = e$
- (D) $a * e = a$

Q60. [1-NAT] [1 mark]

Limit of $\sin(x)/x$ as $x \rightarrow 0$:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-NAT] [1 mark]

Order of $y'' + 3y' + 2y = 0$:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-NAT] [1 mark]

Cauchy sequence:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Union of open sets is:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Normal distribution PDF:

- (A) $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$
- (B) $\exp(-x)$
- (C) $1/x$
- (D) x^n

Q65. [1-MCQ] [1 mark]

$f(x) = |x|$ at $x=0$:

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	D	1
2	1MCQ	C	1
3	1MCQ	A	1
4	1MCQ	A	1
5	1NAT	B	1
6	1NAT	C	1
7	1NAT	B	1
8	1NAT	A	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	A	1
13	1NAT	A	1

14	1MCQ	C	1
15	1NAT	A	1
16	1MCQ	A	1
17	1NAT	A	1
18	1MCQ	A	1
19	1NAT	A	1
20	1NAT	C	1
21	1MCQ	A	1
22	1NAT	A	1
23	1NAT	A	1
24	1MCQ	C	1
25	1MCQ	B	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1NAT	C	1

29	1MCQ	B	1
30	1NAT	A	1
31	2NAT	A	2
32	2NAT	B	2
33	2MCQ	A	2
34	2NAT	B	2
35	2NAT	C	2
36	2MCQ	A	2
37	2NAT	B	2
38	2MSQ	A	2
39	2MCQ	A	2
40	2MCQ	A	2
41	2MSQ	A	2
42	2NAT	A	2
43	2MCQ	B	2

44	2MCQ	A	2
45	2MCQ	A	2
46	2MCQ	A	2
47	2MCQ	B	2
48	2NAT	A	2
49	2NAT	A	2
50	2MCQ	A	2
51	2NAT	A	2
52	2MCQ	B	2
53	2NAT	B	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	C	2

59	2MSQ	A	2
60	2NAT	A	2
61	2NAT	B	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2MCQ	B	2
65	2MCQ	C	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Correct Answer: D

Subject: Algebra | Topic: Linear Algebra | Difficulty: moderate

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Algebra | Topic: Ring Theory | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Matrix | Difficulty: difficult

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Algebra | Topic: Group Theory | Difficulty: moderate

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Calculus | Topic: Real Analysis | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Calculus | Topic: ODE | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Analysis | Topic: Metric Spaces | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Topology | Topic: Open Sets | Difficulty: easy

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Probability | Topic: Distributions | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Real Analysis | Topic: Continuity | Difficulty: difficult

Q11.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q12.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3+1/x^3 = 27-3*3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q13.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q14.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q16.

Answer: A

Solution:

$$L\{t^*f(t)\} = -d/ds(F(s)). \quad F(s) = 1/(s+2). \quad -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q17.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q18.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q21.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q22.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q23.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$L\{e^{-(at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x \cdot \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: A

Solution:

Row 2 = $2 \cdot$ Row 1, Row 3 = $3 \cdot$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q40.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q41.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q42.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q43.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q45.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q46.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q47.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

$$\text{Poisson: mean} = \lambda = \text{variance. Both} = 4.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q50.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q51.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q52.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q54.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q55.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q56.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q58.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q60.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q61.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q62.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3+1/x^3 = 27-3*3 = 18$.

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q63.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q65.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

- Engineering Mathematics: 37 questions, 57 marks
- Miscellaneous: 18 questions, 33 marks
- Algebra: 4 questions, 4 marks
- Calculus: 2 questions, 2 marks
- Analysis: 1 questions, 1 marks
- Topology: 1 questions, 1 marks
- Probability: 1 questions, 1 marks
- Real Analysis: 1 questions, 1 marks

Difficulty Distribution:

- Easy: 14 | Moderate: 35 | Difficult: 16

Statistics:

- Total Questions: 65
- Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.